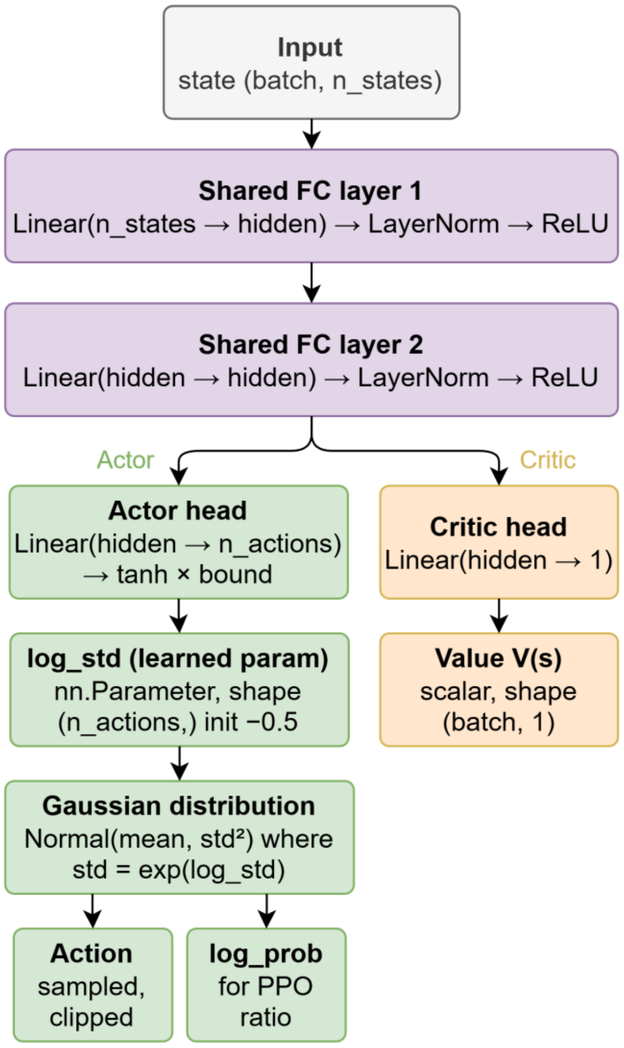
Reinforcement Learning
Algorithms and Applications

Chapter 12-2. PPO for Mountain Car Continuous

PPO (Continuous Version)

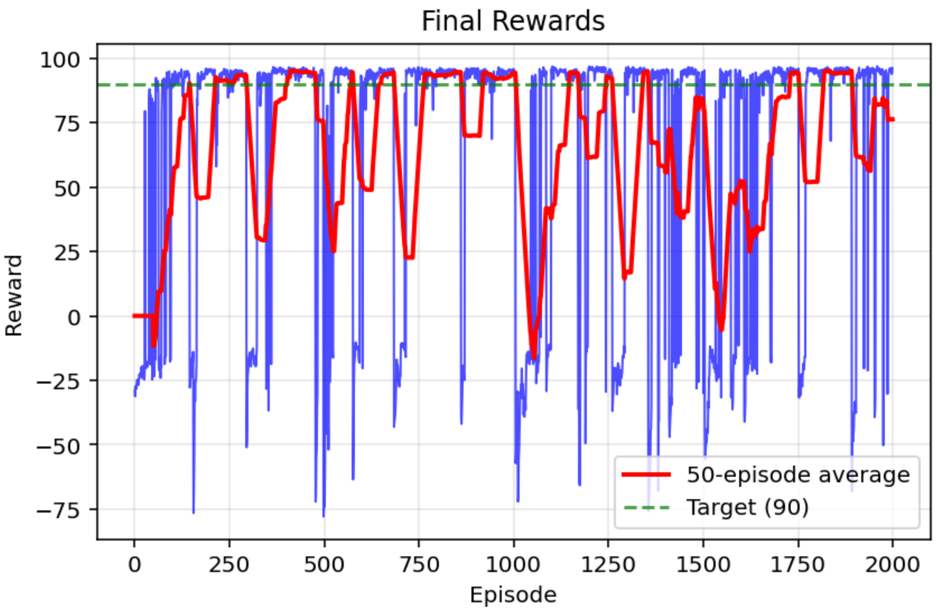
- **State-Independent Standard Deviation**
  - A single global standard deviation is shared across all states.
  - The standard deviation is still learnable during training.
  - The policy outputs only the action mean.
  - Simpler and more stable for training.
  - $\pi(a \mid s) = \mathcal{N}(\mu(s), \sigma^2)$
- **State-Dependent Standard Deviation**
  - The standard deviation is generated from the neural network and varies with the state.
  - Both the mean and standard deviation are learnable.
  - Provides adaptive exploration but may reduce training stability.
  - $\pi(a \mid s) = \mathcal{N}(\mu(s), \sigma(s)^2)$
- **Discrete Entropy**
  - $H(\pi) = -\sum_a \pi\,(a \mid s) \log \pi\,(a \mid s)$
- **Continuous Entropy**
  - $H(\pi) = \mathbb{E}[-\log \pi\,(a \mid s)]$
  - Closed-form for Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$
    - $H(\pi) = \frac{1}{2}\log(2\pi e \sigma^2)$

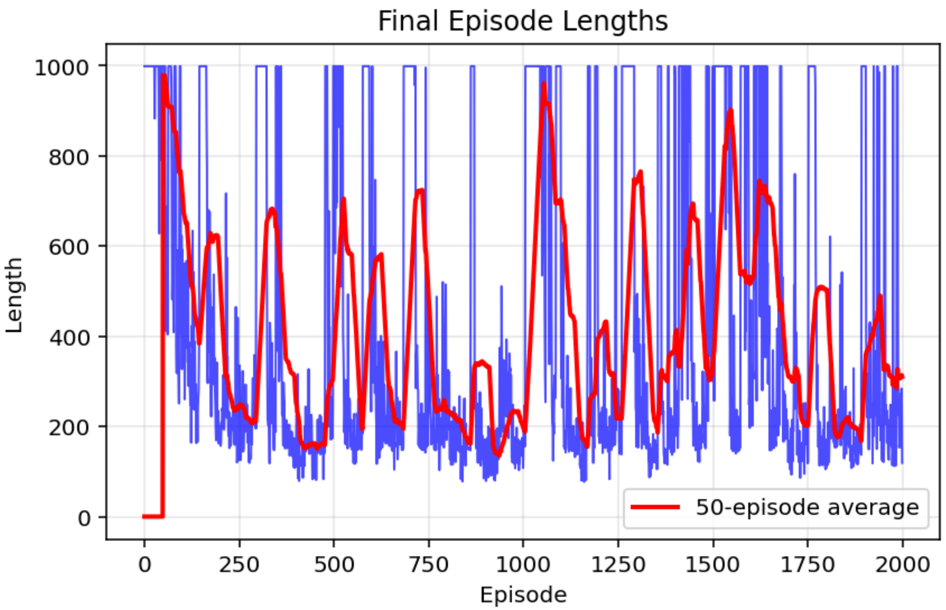


Entropy of a Gaussian Policy

- **Assume the policy is a 1D Gaussian distribution:**
  - $\pi(a \mid s) = \mathcal{N}(\mu, \sigma^2)$
- **The entropy of a continuous policy is defined as:**
  - $H(\pi) = \mathbb{E}_{a \sim \pi}[-\log \pi\,(a \mid s)]$
- **Step 1: Gaussian Probability Density Function**
  - The Gaussian probability density function (PDF) is:
  - $\pi(a \mid s) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(a-\mu)^2}{2\sigma^2}\right)$
- **Step 2: Take the Logarithm**
  - $\log \pi\,(a \mid s) = \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(a-\mu)^2}{2\sigma^2}\right)\right]$
  - Using logarithm properties:
  - $\log \pi\,(a \mid s) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(a-\mu)^2}{2\sigma^2}$
  - Therefore:
  - $-\log \pi\,(a \mid s) = \frac{1}{2}\log(2\pi\sigma^2) + \frac{(a-\mu)^2}{2\sigma^2}$
- **Step 3: Substitute into the Entropy Definition**
  - $H(\pi) = \mathbb{E}\left[\frac{1}{2}\log(2\pi\sigma^2) + \frac{(a-\mu)^2}{2\sigma^2}\right]$
  - Split the expectation:
  - $H(\pi) = \mathbb{E}\left[\frac{1}{2}\log(2\pi\sigma^2)\right] + \mathbb{E}\left[\frac{(a-\mu)^2}{2\sigma^2}\right]$
- **Step 4: Simplify Each Term**
  - The first term is constant:
  - $\mathbb{E}\left[\frac{1}{2}\log(2\pi\sigma^2)\right] = \frac{1}{2}\log(2\pi\sigma^2)$
  - For the second term:
  - $\mathbb{E}\left[\frac{(a-\mu)^2}{2\sigma^2}\right] = \frac{1}{2\sigma^2}\mathbb{E}[(a-\mu)^2]$
  - Since the variance of a Gaussian is:
  - $\mathbb{E}[(a-\mu)^2] = \sigma^2$
  - we obtain:
  - $\frac{1}{2\sigma^2}\sigma^2 = \frac{1}{2}$
- **Step 5: Final Result**
  - $H(\pi) = \frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2}$
  - Using:
  - $\frac{1}{2} = \frac{1}{2}\log(e)$
  - we get:
  - $H(\pi) = \frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2}\log(e)$
  - Combine the logarithms:
  - $H(\pi) = \frac{1}{2}\log(2\pi e\sigma^2)$

Training Result





Reference

- M. Lapan, *Deep Reinforcement Learning Hands-On*, 2nd ed. Birmingham, UK: Packt Publishing, 2020.
- A. Zai and B. Brown, *Deep Reinforcement Learning in Action*. Shelter Island, NY: Manning Publications, 2020.
- S. Ravichandiran, *Deep Reinforcement Learning with Python*, 2nd ed. Birmingham, UK: Packt Publishing, 2020.